

VAMSI KRISHNA KOLLIPARA

(812) 223-8818 | kolliparavamsikrishna80@gmail.com | [linkedin.com/in/vamsikollipara](https://www.linkedin.com/in/vamsikollipara) | github.com/vamsi876

SUMMARY

Forward-deployed engineer shipping LLM systems and Python backends into regulated healthcare environments. Founding engineer on a HIPAA-aligned clinical audit platform — Python orchestration with schema-validated LLM output, delegated authorization, and idempotent ELT on GCP — owned from problem definition through production support. Background in distributed data systems: event-driven services on Kafka, fault-tolerant batch orchestration, and low-latency access over multi-million-row datasets.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript

LLM Systems: Anthropic Claude API, prompt caching, JSON Schema structured output, RAG (Pinecone), LoRA fine-tuning, vLLM self-hosting, human-in-the-loop evaluation

Distributed Systems & Data: Kafka (idempotent consumers), batch orchestration (Google Batch), idempotent ELT, PostgreSQL (Row-Level Security, indexing), Redis caching, asyncio/aiohttp

Cloud & Infra: GCP, Terraform, Azure AD SSO, GitHub Actions CI/CD, Vercel

APIs & Frontend: REST, GraphQL (PostGraphile), React, Next.js, Node.js, Tailwind CSS

EXPERIENCE

Forward-Deployed Engineer | AI Data Management · Remote

Jul 2025 – Present

- Built the Python orchestration layer running a 120-question clinical audit protocol across 6 question-type handlers with JSON-schema-validated LLM output; tripled throughput from 10 to 30+ audits/day while holding 94% agreement against a physician-reviewed gold standard.
- Implemented HIPAA technical safeguards at the application layer: access control, audit logging, and encryption in transit and at rest.
- Designed delegated, scope-limited authorization: Azure AD SSO, centralized RBAC across 8 roles, database-enforced Row-Level Security, and JWT claims propagated into Postgres sessions so authorization is enforced in exactly one layer.
- Operated the LLM pipeline in production: concurrency management within Anthropic API rate limits, prompt caching on shared protocol prefixes to reduce inference cost, and human-in-the-loop review queues keeping every model decision traceable to a reviewer.
- Built idempotent ELT on GCP (Google Batch, Terraform-provisioned): EHR extracts upserted on encounter IDs, so retries and partial failures never duplicate records.
- Shipped the product surface: React frontend over a PostGraphile GraphQL API on Postgres.

Graduate Assistant, Software Engineering | Indiana State University · Terre Haute, IN

Aug 2023 – May 2025

- Built a RAG knowledge assistant over an 8,000-document corpus in Pinecone; tuned embedding models and chunk overlap to improve recall@k for the internal knowledge assistant.
- Fine-tuned LoRA adapters and self-hosted open-source models with vLLM to satisfy campus data-privacy rules prohibiting external API calls.
- Co-authored WebWeaver (published on PyPI), an async Python crawling library (asyncio/aiohttp) with robots.txt compliance, deduplication, and BFS/recursive modes; used it to ingest 40,000+ university URLs and distill the 8,000-document policy corpus.

Programmer Analyst, Full-Stack | Cognizant (Client: Duke Energy) · Hyderabad, India

Jan 2022 – Jul 2023

- Built REST services serving metering and billing data to a customer platform with 10,000+ daily active users.
- Cut p95 latency from 2.5s to under 1s on multi-million-row smart-meter (AMI) tables via composite indexing and query optimization.
- Added Redis read-through caching (15-minute TTL) on read-heavy metering endpoints to absorb peak traffic.
- Implemented Kafka consumers for outage and billing notification workflows with idempotent handling — retries are safe and customers are never double-notified.

OPEN SOURCE

- WebWeaver** (PyPI, Sep 2024) — async web-crawling library with robots.txt compliance, deduplication, and BFS/recursive modes. Python, asyncio, aiohttp.
- GitBar** (Mar 2026) — macOS menubar dashboard aggregating PRs, issues, and CI status across GitHub, GitLab, and Bitbucket. Python, PyObjC, REST APIs.

EDUCATION

- M.S. Computer Science**, Indiana State University — May 2025 — GPA 3.66
- B.Tech, Electronics & Communication Engineering**, Vel Tech University — May 2022 — GPA 9.01/10